

3(a)	<i>any two from</i> 1 (division that) produces, new / daughter, cells A produces more cells or (so) replaces, damaged / lost / dead, cells ; 2 new / daughter, cells, genetically identical ; A <i>ref. to</i> clone if correct context A genetic information not lost I same number of chromosomes 3 (all) new cells can retain function ; AW	2
3(b)	<i>any two from</i> 1 cell cycle continuous / continually divide / AW ; 2 (produce cells that) can, differentiate / specialise / described ; A multipotent / pluripotent I totipotent <i>divide to produce a cell that can divide and a cell that differentiates = 2 marks</i> 3 can produce, cells / tissue, that can still function (as before) ;	2

Question	Answer	Marks
3(c)	<i>any three from</i> 1 chemicals (released) are cell signalling, molecules / compounds ; 2 liver cells, are target cells ; A liver tissue for liver cells 3 binding of, chemicals / cell signalling molecules, to receptors (of liver cells) ; R receptor cells 4 <i>ref. to specificity (of receptors) / chemicals complementary to receptors ; if R above, then allow ecf for idea of complementary</i> 5 (specific) response is, cell enters the cell cycle / mitosis / cell division ; A DNA replication 6 AVP ; e.g. <i>idea of communication between cells</i> suggestion of detail following binding, e.g. second messenger activated / enzyme cascade /signal transduction / phosphorylation events / enzyme activation I cascade of reactions	3